

SLA or Nay?

The impact of selective licensing on rental market outcomes in Bristol

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The private rental sector is a massive and important part of the society and economy of any nation. The private rental sector's role as a major source of speculative investment, in addition to its critical role as a necessary good for human survival, drives intense interest in the contemporary structure and regulation of the private rental sector. As a consequence, regulation of the private rental sector is controversial. This paper examines one such regulation strategy in the United Kingdom, discretionary licensing strategies, which require minimum quality standards for rental properties in certain areas or that fit certain characteristics. While study of local housing policy is rare, academic study of private rental sector regulations *beyond* price controls are rarer still. Hence, this paper seeks to build understanding about the specific short-run impacts of the adoption of discretionary licensing policies in Bristol from 2019 to 2024 using spatial causal inference techniques. I find that there is no significant impact on market clearing, while the licensing scheme has a moderating effect on rent level growth at the boundary. This suggests that private rental sector regulation targeting minimum quality standards, such as those guaranteeing minimum energy efficiency, gas and electric safety, and other quality inspection checks, may serve to cool off rental markets, increasing clearing speed while reducing the average rate of rent growth.

Introduction

The private rental sector is a large and significant factor in many countries' economies and societies. Rental markets define individuals' lives, in some senses; they govern where most people in their 20s live, and increasingly they control how and where people are able to work and start families. As a consequence, the regulation of the private rental sector is a significant political issue. While most analyses of rent control policies suggest that price-based rent controls have mixed effectiveness when considered as a whole across the city (Kholodilin 2024), especially in their effects on the quality of privately-rented properties. However, there is not a clear agreement in the economic and policy studies literature on the impact of *minimum*

quality standards regulation on the rental market; grey literature specifically focused on the UK finds relative success in the adoption of quality and licensing standards (Ministry of Housing 2019), while other political advocacy groups argue that this is incorrect and ignores potential long-term and secondary effects of regulation of the private rental sector. Further, many of the policy models in the United Kingdom target specific segments of the private rental sector, such as rental properties with more than 5 people that comprise at least two separate households, or properties that fall into specific urban wards. These kinds of neighborhood-level rent regulations are less-well studied (O’Toole 2023), and the evidentiary requirements for understanding their impacts or justifying their use legally tend to be quite challenging to satisfy fully (Gibb and Marsh 2022).

This paper seeks to shed light on this issue through a unique adoption pattern of neighborhood housing quality regulation policies in Bristol. Bristol City Council has adopted private rental sector regulations, known as *discretionary licencing* policies, that apply to specific urban wards. These policies, as discussed below, have different impacts on different types of properties, and may generate localized externalities that distort local urban rental markets. This paper seeks to identify whether, specifically, covering most of central Bristol in discretionary licensing areas from 2019 to 2024 introduced any specific short-term effects on two important private rental sector outcomes: rent levels and turnover. This seeks to complement previous reviews of neighborhood-level rental policy with additional evidence using a novel dataset on private rental sector listings, a unique causal identification strategy, and two separate estimation strategies.

Background: What are DLAs?

The *discretionary licensing area* is a sub-city private rental sector regulatory tool intended to tackle specific urban social problems that are seen as relating to or otherwise influenced by issues in the housing market. Selective licensing areas arise from the *discretionary licensing* authority provided by the UK government to local authorities. This places a few requirements on how and why discretionary licensing policies can be enacted. For our purposes, discretionary licensing policies can be enacted to resolve issues such as low housing demand (i.e. urban blight) anti-social behavior, low housing supply, or poor quality housing, but also can include areas of high migration. As such, DLAs must be specific, targeted neighborhood-level regulations that seek to achieve a specific policy outcome, such as increasing housing demand as necessary to reduce urban blight, reduction in anti-social behavior ordinances or general crime, improvement in housing conditions, or reducing (or limiting the growth of) overcrowding due to in-migration. Further, DLAs are time-limited, only lasting five years with the possibility of renewal. *Selective Licensing*, which applies to all properties on the private rental market, can operate alongside *additional licensing* schemes that target specific segments of the rental market. In many UK cities, additional licensing requirements apply to homes of multiple occupation *HMO*—dwellings where tenancies involve more than two people sharing a toilet, bathroom, or kitchen—and typically involve documentation and/or inspection requirements for the property.

In Bristol selective and additional licensing requirements have been applied across the city at different times. As a neighborhood-level rental market regulation, two city wards (Eastville and St, George West) have been designated as selective licensing areas (SLAs) from 2016-2021, but a very large portion of Bristol was covered with additional licensing regulation starting in 2019 and ending in 2024. This included most of the wards in central Bristol, as shown in Figure 1. In Bristol, selective licensing requirements are inclusive of additional requirements, meaning that the introduction of *selective* licensing includes all properties that would be captured in an additional licensing scheme. We denote this as *selective+* for specificity, while additional licensing will be referred to as such. Treatment under any DLA scheme occurs in three tranches; the first applying selective licensing to all private rental properties in two eastern wards, the second applying additional licensing across central Bristol, and the third applying selective licensing in two south Bristol wards and extending additional licensing northward.

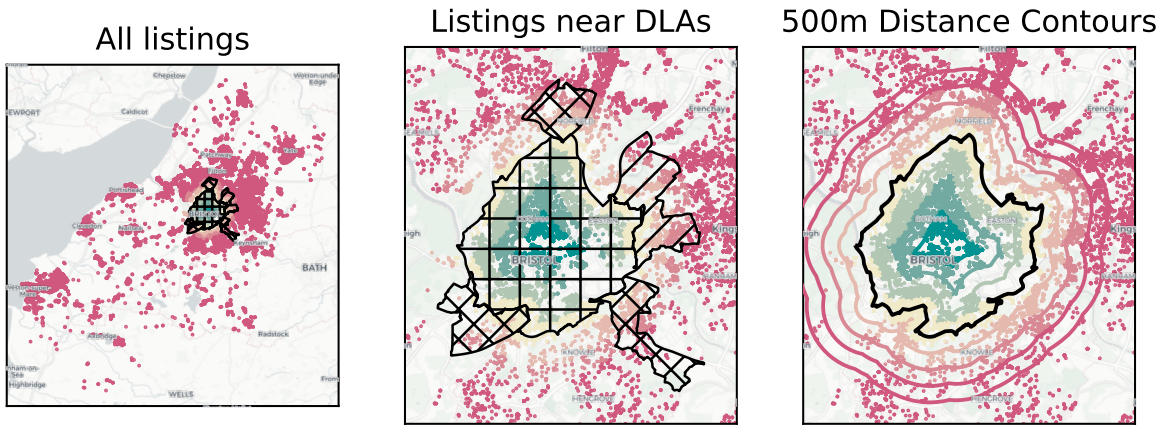


Figure 1: Map of rental listings in Bristol, as well as selective (X and //) and additional (+) licensing area boundaries.

Table 1: Bristol wards included in tranches of selective and additional licensing policy introductions.

	Ward	Type	Tranche	Start	End
0	Eastville	Selective+	1	2016-07-01	2021-06-30
1	St George West	Selective+	1	2016-07-01	2021-06-30
7	Ashley	Additional	2	2019-06-10	2024-07-07
8	Bishopston & Ashley Down	Additional	2	2019-06-10	2024-07-07
4	Central	Additional	2	2019-06-10	2024-07-07
10	Clifton	Additional	2	2019-06-10	2024-07-07
11	Clifton Down	Additional	2	2019-06-10	2024-07-07
9	Cotham	Additional	2	2019-06-10	2024-07-07
5	Easton	Additional	2	2019-06-10	2024-07-07

Table 1: Bristol wards included in tranches of selective and additional licensing policy introductions.

	Ward	Type	Tranche	Start	End
6	Hotwells & Harbourside	Additional	2	2019-06-10	2024-07-07
3	Lawrence Hill	Additional	2	2019-06-10	2024-07-07
13	Redland	Additional	2	2019-06-10	2024-07-07
12	Southville	Additional	2	2019-06-10	2024-07-07
2	Windmill Hill	Additional	2	2019-06-10	2024-07-07
16	Bedminster	Selective+	3	2022-04-06	2027-04-05
15	Brislington West	Selective+	3	2022-04-06	2027-04-05
14	Horfield	Additional	3	2022-04-06	2027-04-05

Implementation

So, what must be done if a property is regulated under selective or additional licensing designation? Typically, it requires the landlord to pay a fee of between £900 and £1500 pounds (depending on the configuration of the property and the licensing scheme), and formally record property quality information, such as a gas safety record, an up-to-date energy performance certificate, and an electrical installation condition report. Together, these are intended to ensure that the property is of sufficient quality as to be let to tenants.

Methods and Data

In order to estimate these effects precisely, I use a novel dataset of scraped property listings from a private rental sector data provider, vendored through the Urban Big Data Centre at the University of Glasgow, UK. The data ostensibly provides information on all of the private rental properties listed in the UK from 2018 to 2023. I take a subset of this data focused on Bristol post codes, and filter the data for apparent duplicates, such as listings that are made for the same property over the same time period. Thus, this provides fairly solid data spanning the introduction of SLAs in 2019 and 2022, providing information on property characteristics, such as the number of bedrooms, bathrooms, and reception rooms available to rent with the listing, as well as user-contributed pictures, information on the letting agent (if applicable), full-text, short-text, and bullet-point descriptions of the listing. I take as outcomes the weekly rent charged and the duration the listing is live, and I use exogenous information about the property, such as the number of bedrooms, bathrooms, reception areas, etc. as predictors.

Occasionally, a listing’s characteristics will be modified during its lifetime. This might happen, for instance, if a one-bedroom property is listed as a two-bedroom property by mistake. These errors are rare, but I still only retain properties whose primary characteristics do not change

during their listing. Alternatively, changes to a listing’s *price* can indicate that tenants are not expressing interest in the property at the initial price point. Fortunately, the data provider lists the sequence of price changes over the life of the listing (although I do not know when in the listing life the changes are made). From this, I can see that the general trend for price updates over the life of the listing is negative, suggesting landlords or agents are reducing the price in order to find a tenant willing to accept that price. Therefore, I take the *last price* on the listing as indicative of the clearing price.¹ I take the duration the listing was live, from first post to removal from the platform, as an indication of market turnover. Since the measure increases as the turnover slows, I call this the *staleness* of the listing. The platform itself requires the removal of listings that have gone let-agreed, so we can be reasonably sure that the listing length corresponds to time-on-market for most properties, but, some clearly-stale listings stay on the platform for over a year. Thus, I anticipate this outcome to be less straightforward to model in this application, and take the results of this staleness analysis with a grain of salt.

Unfortunately for us, this timeline is also fundamentally confounded by the COVID-19 pandemic. While we can thus make indicative statements of the six-month period after the introduction of the first SLAs pre-COVID, we cannot make hard claims about the specific effect of the SLAs separate from the time over which they were implemented. In essence, this is true of any policy being analyzed, but in practice the COVID-19 pandemic and associated stay-at-home orders were exceptional impacts on life in the United Kingdom (and elsewhere) during the study period. Indeed, these effects also adjusted individuals’ likely tastes and preferences for short-term housing pushing people further out of the city center (Spell and Perry, n.d.).

In effect, this forces an identification strategy of three periods: (A) pre-SLA, (B) the immediate short-term impact of SLAs on the private rental sector in the nine months since first deployment, and the (C) long-run post-lockdown market structure. These periods are clearly evident in Figure 4, Figure 2, and Figure 3. While we can clearly see a reduction in supply across the treatment boundary in Figure 3, the later COVID-19 pandemic clearly suppresses supply further. However, listing geography is quite heterogeneous, as noted in Figure 6; *where* supply decreases (and thus price increases) is almost as important as *that* these changes occur. Hence, we need a more precise spatially-local identification strategy to differentiate these outcomes.

Therefore, when thinking about the identification of the effect of selective licensing areas on various rental market outcomes, like rent levels, rent growth, turnover, or volume, it is important to acknowledge a few things that can improve our ability to assign changes directly to the effect of SLAs. First, aggregate comparisons of rent levels or private rental sector size and turnover are likely to be hopelessly confounded by the wide-reaching changes induced by the COVID-19 pandemic, given the wide-ranging changes in life due to the stay-at-home orders.

¹Note that I do not argue this is actually the rent *charged* to tenants. Many tenants inquiring about rental listings in this period may have engaged in post-listing bidding competitions with other tenants, driving the final rent charged higher than the public market clearing price. Alternatively, the listing’s original post date is visible on the platform, so some savvy tenants may choose to under-bid the public price on a particularly stale listing.

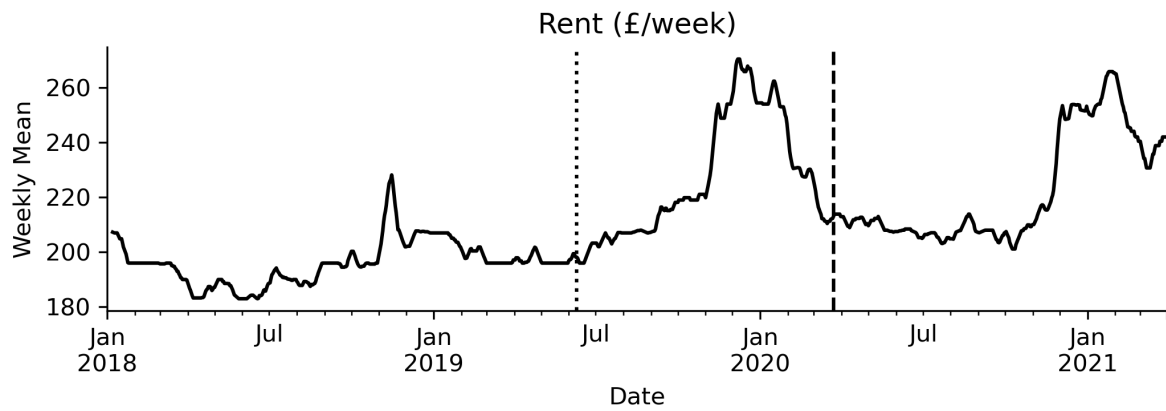


Figure 2: Evolution of the advertised rent for privately-let properties on the rental market database.

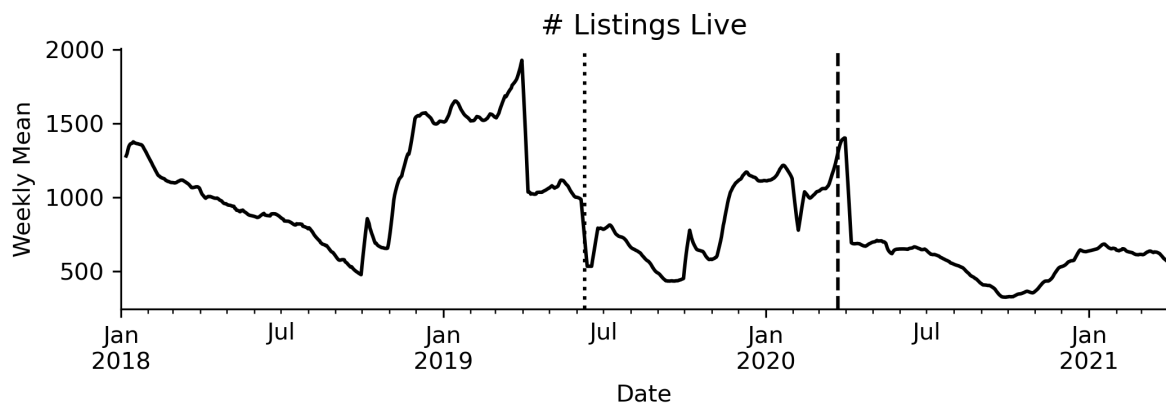


Figure 3: Typical number of listings live within a week on the private rental market database.

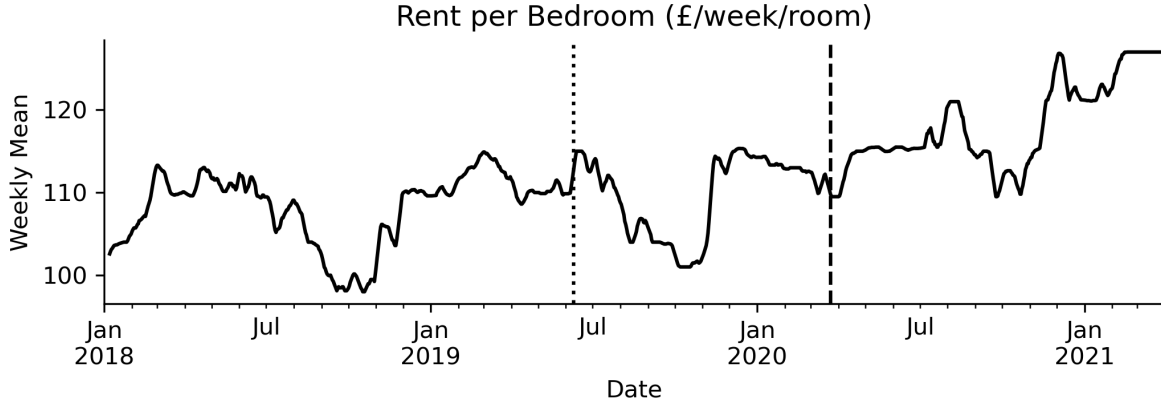


Figure 4: Rent per bedroom for listings on the private rental market database.

Therefore, this paper uses two different strategies in order to create a structured comparison between treated and untreated properties.

First, we will attempt a straightforward spatial difference in discontinuities specification for this data. At the DLA boundary, properties are most likely to be substitutes, while far outside (or inside) the DLA boundary, properties are unlikely to be substitutes—a city center property is never going to reasonably be compared to a property in a far-out commuter community. However, a standard difference in differences application would construct such a comparison. Therefore, we need to focus on the local effect close to the DLA boundary. To do this, the SDIDc strategy calculates the discontinuity in outcome that exists at the DLA boundary *pre-treatment*. This acknowledges that listings are really only comparable directly near the boundary of the DLA. Then, once the DLA becomes active, we can then estimate this discontinuity again. The difference in these discontinuity estimates reflects the relative contribution of the DLA to the dynamics of price changes in the area local to the boundary. Imagine that the DLA ends up inflating rents through cost passthrough, as claimed by landlord interest groups. If this is the case, then properties near the boundary within the DLA should grow in rent faster than those outside the boundary. If the boundary *starts* showing that properties within the DLA are more expensive, then we expect *the difference in discontinuities* to be positive: the discontinuity between DLA and not-DLA should grow. In truth, we see the exact opposite; properties just inside of the DLA boundary start slightly cheaper and become much cheaper after a year of DLA implementation.

Second, we will use a spatial *X*-learner (Künzel et al. 2019) strategy to address any imbalance in the treated/untreated cohort at the boundary. This is a causal machine learning strategy whereby separate estimators are fit to the treated and untreated groups. Treatment effects are estimated as the difference between the treated model’s predictions and the untreated model’s predictions, weighted by the propensity of a property to enter the treatment group. This second propensity model corrects for potential sample imbalance, since observations just within the

treated area may systematically differ from observations just outside of the treated area. Like typical doubly-robust estimation, the causal effect is identified when the treatment and control models are well fit or the propensity score model is well-fit. Further, the *X*-learner collapses to a simpler *T*-learner (i.e. two-model learner) if there is perfect separation between treatment and control models. And, it further collapses to an even simpler *S*-learner (i.e. single-model learner) if there is no heterogeneity in the treatment and control models. Given the intensely spatial nature of the rental market and the possibility of spatial heterogeneity in the rental market process, the *X*-learner is necessary for our case. In our specific empirical example, this *X*-learner will be composed of a local regression fit on data within the DLA, a local regression outside of the DLA, and a local logistic regression on data near the boundary. The causal effect is then estimated using the difference between the predicted outcomes for houses on the boundary, weighted by their propensity to be treated.

I expect the *X*-learner to work well in this case given the arguments above. And, for cross-sectional analysis, I expect such a model will work quite well. However, given rental pricing is notoriously *sticky* and experiences a unique ratcheting effect (Wang and Zhou 2000; Gallin and Verbrugge 2019), we also need to control for changes in time across the treatment introduction. For this model to adequately control for starting conditions, we use a unique matching procedure focused on creating a quartet of observations that form a structured comparison between the treated observation and its predecessor pre-treatment and another observation identical to the treated observation (a spatial substitute), which would never be treated. Conditioning on these predecessors, we this analysis shifts the identification strategy to a *spatial difference in discontinuities* analysis: we seek to identify whether there is a discontinuity at the treatment boundary in a model of the outcome *controlling for* the trend.

Still, simply comparing the differences between these matches' differences over the policy boundary is only a partial story—the actual policy areas where DLAs are implemented are decidedly non-random. In some sense, it is impossible to compare one property in the center of Bristol to another at the periphery and assume that their rental dynamics will (or ought) to be the same. The dynamics of city-center properties will nearly always differ from city-edge properties, hence the importance of the spatial component of the identification strategy.

Finding Local Spatial Substitutes

We consider the use of treated *local spatial substitutes*, or nearby similar properties, to pre-treatment listings. These can be thought of as approximate matches, in the sense that one two-bed one-bathroom end of terrace might be judged as a substitute for a similar property in the same estate listed later. Indeed, matching locally to treated spatial substitutes is a useful technique to ensure that approximate comparisons can be made across the data, maximizing the use of the initial pre-treatment listings. Notably, this is quite different from simply picking later matched properties that are most similar on attributes—indeed, spatially-constraining the matches will invariably *degrade* any a-spatial measure of match quality if there are any non-local optimal matches. However, using *local spatial substitutes* in place of exact nonlocal matches

ensures that substitutes have (nearly) the same set of unmeasured locational confounders as the target property for matching.

To clearly define the matching procedure, it helps to define a few different types of observations. First, a *treated* listing is one made in a DLA that was made during the treatment period. I distinguish *pre-treatment* listings made in a ward *before* the DLA came into effect, while a *never-treated* listing is one located in a ward where a DLA was never introduced. A *substitute* is a donor listing near (or colocated with) a target listing with identical characteristics. If substitutes are *contemporaneous*, they are live at the same time, such that a prospective tenant might actually have considered both properties. Thus, we make pairs of treated listings and untreated substitutes for geographic matches. Then, we find a listing made before the DLA came into effect, called a *precedent*. Then, we aim to construct a *quartet* of observations, all with identical characteristics, such that the treatment listing and its predecessor, the treatment listing and its substitute control, and the substitute control listing and *its* predecessor are close to one another.

This can be optimised in two ways. First, it helps to note that finding a precedent for a treated observation *does not* affect finding a substitute, since substitutes are outside the DLA and the pretreatment precedent is inside the DLA. Therefore, the treated-pretreatment matching can be optimised separately from any other decision. Second, the match quality for selecting a substitute for each treated observation *must* consider both the distance between the treated observation and its substitute *and* the substitute and its precedent. For a simple example, imagine we have two potential substitutes available for treated observation *A*: substitute *B* is 250 meters away and substitute *C* is 300 meters away. If we choose substitute *B*, we then must find a precedent for *B*. If *B* has no nearby precedent, then we might instead prefer selecting substitute *C*. Indeed, in the worst case, *C*'s predecessor *C'* is colocated with *C*, so the total match distance from assigning *C* as *A*'s substitute is just the distance between *A* and *C*, $d(A, C)$. However, *B*'s only available precedent might be *C'*, making the total match distance from assigning *B* as *A*'s substitute $d(A, B) + d(B, C')$. Hence, once there are at least two substitutes, we must consider the choice of substitute *and* the substitute's precedent at the same time in order to optimize the entire quartet. Simply optimizing each step in turn may lead to suboptimal outcomes. Finally, consider that while we might seek matches for all treated observations, we only really need to consider spatial substitutes near the boundary of the DLA for the purposes of the analysis. Thus, including all treated properties in the matching will likely degrade the performance of the matching procedure, as interior treated observations consume available untreated candidates for matching. Therefore, we will only construct quartets for treated observations within 2 kilometers of the treatment boundary.

	uid	Treated	In DLA?	Date	Postcode	Beds	Baths	Rent
kind								
treated	9cae7e1edb	True	True	2019-08-14	BS34AP	2	0	242
pretreat	6f91be8c67	False	True	2018-08-16	BS34AP	2	0	230
substitute	573f5484f2	False	False	2019-06-03	BS44JD	2	0	207

	uid	Treated	In DLA?	Date	Postcode	Beds	Baths	Rent
kind								
presubstitute	70cfae8c52	False	False	2018-01-07	BS44AU	2	0	167

Identification Challenges

Finally, it is useful to discuss a few challenges to identification.

First, properties listed in 2018 and again after 2019 may not necessarily be typical properties for the private rental market. That is, if a property is re-let every year, it’s unlikely to reflect long-term stable tenancies that, while not common, are nonetheless present. Indeed, DPS estimates place the typical tenancy close to three years (Service 2025). Unfortunately, given the span of the data, we cannot significantly lengthen the study window to capture this effect. However, this does magnify the importance of the local spatial substitutes—since any one property let in 2018 may *never* recur in our data, we can leverage nearby similar properties as a proxy to understand what such a property would obtain if re-let on the private rental market at a given point in time. This is a secondary motivation for using spatial substitutes to define predecessors, if necessary.

Second, we make no specific assignation of whether the DLA causes changes in the housing market through the landlord or the tenant mechanism. To illustrate, renters might view the DLA as conferring extra protections (it does not necessarily do so) or obligations on the landlord to the tenant (similarly so). If they do, any premium for regulated properties might arise simply from renters *preferring to live in licensed properties and, as such, live within a DLA*. Alternatively, any supply or turnover changes may similarly arise from altered tenant behavior. A tenant on the margin such might attempt to extend their tenancy instead of finding new accommodation if they valued its status as a licensed property. On balance, this would result in higher local rents and slower local rental markets with fewer properties being let per unit time. Even if there is only an *additional* licensing scheme in a given DLA, this effect may occur as pressure on HMO supply, driving tenants into smaller less-affordable households with fewer co-tenants.

The supply-side story is quite different, but would have similar apparent empirical effects. Given that discretionary licensing compliance is required regardless of whether or not a tenancy turns over, it would not specifically incentivise landlords to retain (or turf out) tenants. However, landlords still might view obeying the terms of the DLA as an onerous *extra cost* and thus seek to exit the rental market, since there is both a price of applying for a license, certification processes to ensure minimum quality for rented accommodation, and (potentially) an inspection by the city council during the term of the license (typically five years). This combination of fiduciary and non-fiduciary costs could indeed depress landlord participation in the rental market on the margin, which would reduce supply. In turn, this mechanism then may increase rents charged and lower market turnover. This latter justification was decisively argued by

community groups during the local consultations about DLA implementation, as well as at least one UK-wide landlord advocacy group.

In order to address questions about identifying the specific mechanism by which changes affect the market, we would need more precise information about the rental market, such as within-tenancy rent changes and links to house sales to confirm rental market exits. The latter is possible, but the former is unlikely without an officially-maintained private rental sector database.

Estimation

Both the SDIDc and X-learner identification strategies construct their causal effect estimates by comparing *predictions* between actual and counterfactual treatment statuses. This means we *do not* necessarily need to focus on a specific parameter estimate for the control variates. Indeed, this liberates us to use quite flexible local models. Thus, we use geographical random forests (Georganos et al. 2019) from the `gwlearn` package (Martin Fleischmann et al. 2026) to construct locally-nonlinear models relating exogenous listing-specific predictors (like the number of bedrooms, bathrooms, reception areas, floors) to the outcomes under study. I allow for outcomes in one model to predict the other model, as well, meaning that the model analyzing rent levels considers listing duration as a predictor, and the model analyzing staleness can consider the advertised rent as a predictor. I use simple 5-fold crossvalidation to train the random forests, and use the same approach to identify a scale parameter that governs the local model.

For the spatial difference in discontinuity model, we estimate one model for each treatment status (pre/post and in/out of DLA) and outcome. For the X-learner, we estimate two models for each condition and outcome (one for treatment and another for control), and one propensity model used for both. The first model is a predictive model of the outcome. The second model is used to *impute* potential outcomes across the treatment category.

For the SDIDc model, we compare the predicted difference at the boundary pre-treatment to the predicted treatment at the boundary post-treatment. Therefore, pre-treatment control is unnecessary directly in the post-treatment models. However, the X-learner does use pre-treatment controls in each model. This means that we include predecessor listing outcomes (rent or listing duration) as covariates for the post-treatment prediction.² Then, we calculate predictions for each observation’s *actual* model—the model trained on data that matches the observation’s treatment status—and compare this to the prediction from that observation’s

²I acknowledge this is slightly imperfect, since any two quartets may have different delays between current and predecessor listings; if one listing in 2020 might be matched to a predecessor in 2018, while a different listing in 2022 might also be matched to a predecessor in 2018. While the rate of rent growth may be constant, the two will experience quite a different relationship between previous and current rent. I am currently re-estimating the X-learner by converting the outcome to the effective annualized rate of change for outcomes in each quartet in/out of the DLA.

counterfactual model, trained on observations that have a different treatment status. The propensity score for X-learners uses all available data to predict the likelihood of treatment.

Results

In terms of describing the matching process and data characteristics, it's useful to note a few subtotals. Around half (20,238) of the 42,838 listings are in treated areas, with the vast majority of potentially treated listings (16,198) occurring in the Bristol additional licensing area adopted from 2019 to 2024. Out of these listings in the central DLA, about half (7,637) occur after the introduction of the DLA, half again are within 1km of the boundary (4,581). Of these listings, a simple matching that minimizes the distance separating matched properties with identical characteristics manages 93% efficiency: 4,293 listings are fully matched. The procedure finds predecessors with identical characteristics let at least a year prior that minimize the total distance from treated and predecessor listings. Then, the procedure finds the optimal set of contemporaneous untreated substitutes (within 30 days), and then optimal substitute predecessors let at least a year prior. Because the matching is exact, no listing is used twice.

The locality of these matches still leaves something to be desired. Maps and match distance distributions are shown in Figure 5. While we might expect short match distances within the city, it's not obvious why substitutes are drawn linking, say, listings in the southwest outskirts of Bristol (i.e. Long Ashton) out to dormitory towns (Clevedon, Nailsea); while these are indeed substitutes for certain kinds of tenants, they are not as hyper-local as one might hope. Longer-term temporal data may improve this, but the analysis is also fundamentally limited by the size of the rental market in Bristol.

Regardless, the covariate balance is exceptional for these quartets: distributions are shown in Figure 6, and confirm that the exogenous covariate match dimensions (beds & bathrooms) match perfectly and uncontrolled exogenous dimensions (reception count) are also undistinguishable among matches, while endogenous values (rent charged, listing time on market) differ between some quartet components. This indicates the matching has worked as intended: drawing a property at random, we couldn't distinguish whether it was a treatment listing, a predecessor of a treatment listing, a substitute, or a predecessor of a substitute unless we looked at its *outcome* values. Parametric tests of this balance are shown in Table 3, which provides two-sample Kolmogorov-Smirnov tests across pairs of the four groups along five covariates in Figure 6, largely confirming the visual intuition.

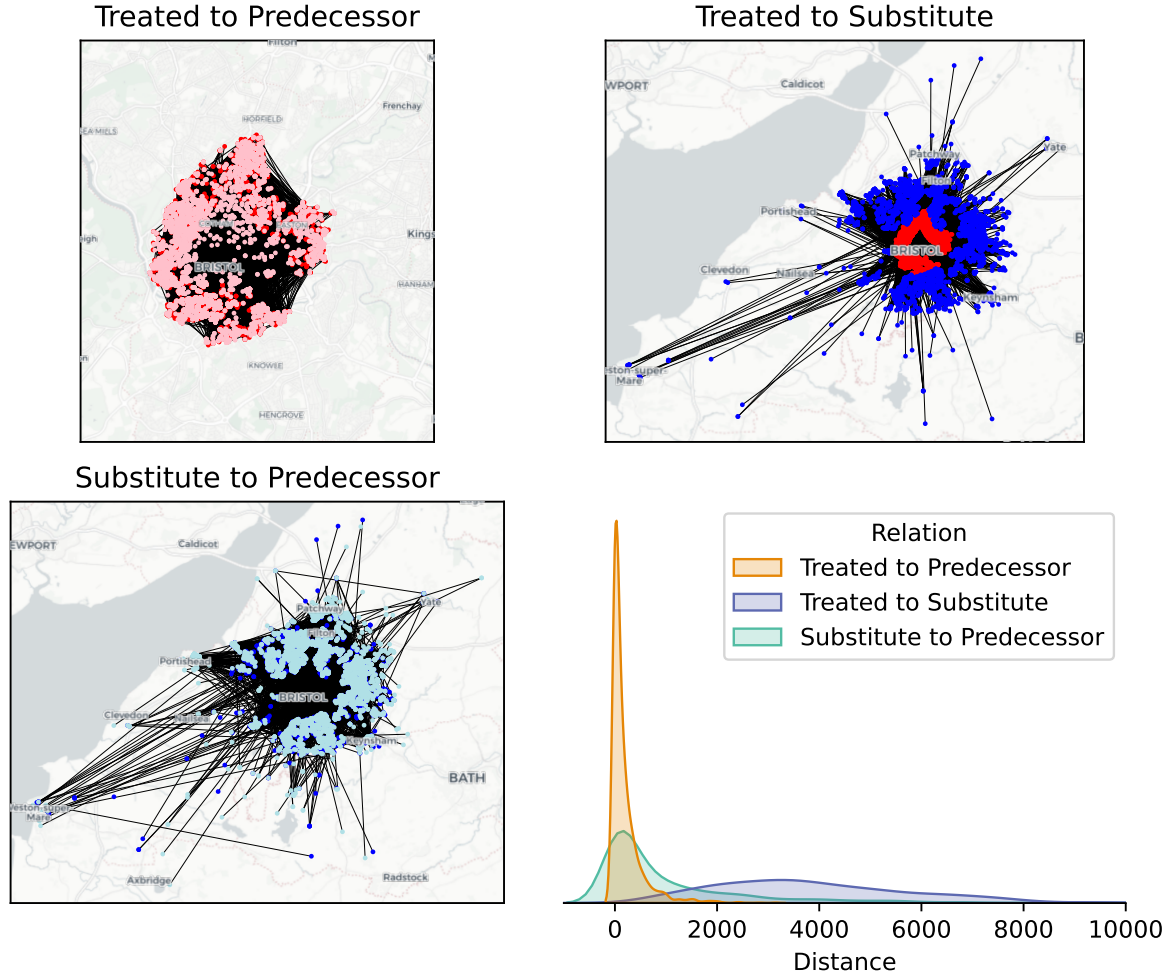


Figure 5: distribution of treated listings near the treatment boundary and their precedents, as well as substitutes and their precedents, and the distances between treated observations and their precedent, treated observations and their substitute, and substitutes and their precedent.

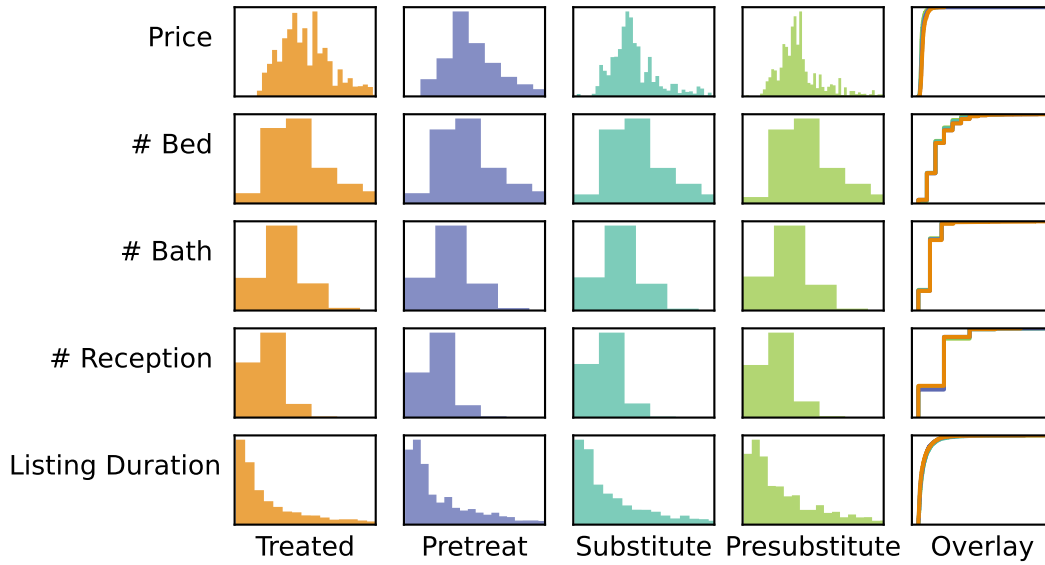
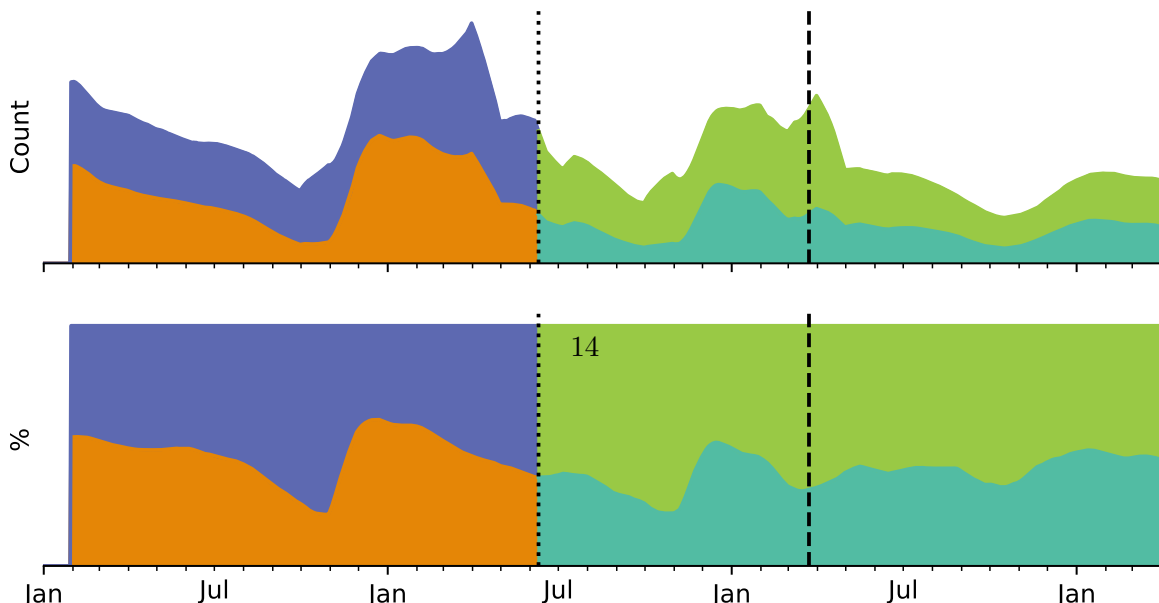


Figure 6: Distributions of relevant covariates for listings in each part of a quartet.

Table 3: Parametric pairwise tests for distribution balance using Kolmogorov-Smirnov tests with a conservative Bonferroni correction for multiple comparison. The distribution of each variate is tested across pairs within the quartet; significant results indicate that two quartet components' distributions may differ.

	Variable	Quartet A	Quartet B	KS Stat	p-value	variable
0	price_last	treated	pretreat	0.11	0.0	Price
1	price_last	treated	substitute	0.25	0.0	Price
2	price_last	treated	presubstitute	0.34	0.0	Price
3	price_last	pretreat	substitute	0.15	0.0	Price
4	price_last	pretreat	presubstitute	0.26	0.0	Price
5	price_last	substitute	presubstitute	0.12	0.0	Price
25	listing_duration_	treated	substitute	0.06	0.0	Listing Duration
26	listing_duration_	treated	presubstitute	0.08	0.0	Listing Duration
28	listing_duration_	pretreat	presubstitute	0.06	0.0	Listing Duration



over the winter months in 2019 and 2020, as well as the post-COVID dropoff in March of 2020. However, there are no immediately apparent changes to the listing totals, either compositionally or in total across the treatment boundary. One can also see quite a different pattern in Figure 8, where listing durations peak in the summer and decline into the winter. This effect is pronounced for the pretreatment areas within Bristol City Center, and is quite muted for the untreated areas. In addition, we see a slight peak in listing durations immediately after the introduction of DLAs in Bristol city centre, but it is unclear whether this is an effect of the policy.

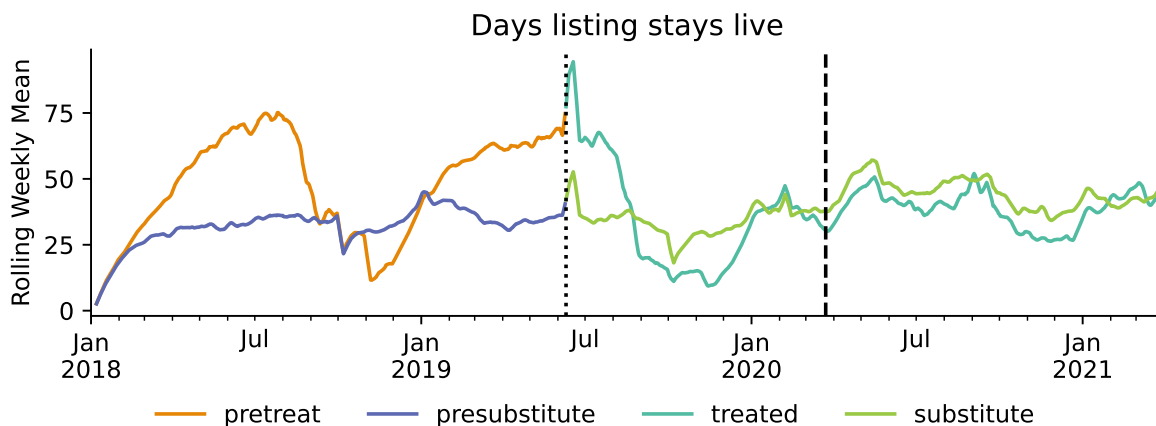


Figure 8: Typical length of time listings are live, calculated as the average duration of listings live within a week of the given date.

Spatial Difference in Discontinuity

In terms of the spatial difference in discontinuity model, we can from Figure 9 see the models are passable at predicting the price across treatment boundaries, but the duration predictions are fairly poor. This comports with our expectations from first principles about what “staleness” represents in this data. Still, the price predictions are passable here, given the heterogeneity of the market and the relatively limited information used to make predictions.

Table 4: Estimated ATE and group discontinuity differences for the SDID specification

	level_0	gross	gross.1	weighted	weighted.1
0	outcome	duration	price	duration	price
1	ATE	1.4137	127.3117	7.3919	60.4241

The group means for the SDID specification are shown in Table 4, while the estimated treatment

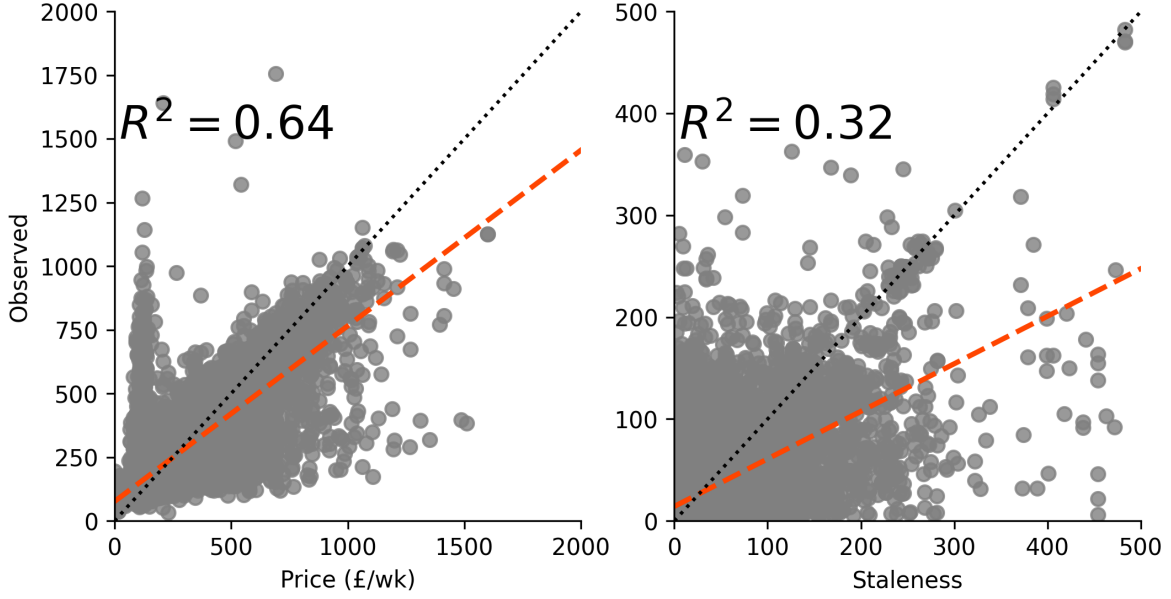


Figure 9: Spatial Difference in Discontinuity residuals, pooled across treatment status

effects are shown in Figure 10, with significance assessment using wild/simple bootstraps. From Table 4, we can see a distinctive difference in the trends inside and outside the DLA.

Before DLA introduction, the discontinuity between models within and outside of treated wards was about £38, or about 2 days. This means that if you predicted the rent of a property using a model within the DLA, your prediction would be lower than one generated from a model outside the DLA by about £38 pounds. This initial discontinuity is a straightforward comparison of the local rents near the boundary pre-treatment. This discontinuity *flipped*, so that models trained in the city center began to predict higher prices while those outside did not. The signed change in discontinuity, then, is about £60; this is the change in the gap between treated and untreated areas after treatment. Hence, we can see that the introduction of the Bristol City Center DLA flipped the rent disparity by about £60 pounds per week, inflating rents inside the boundary (bottom row, right pane). Over the entire regulated area (top row, right pane), the effect was more pronounced (£127 per week), but this comparison includes very central properties for which outlying rentals are not really substitutes. Examining listing duration, however, the DLA had a marginal impact overall, and close to the boundary; turnover tended to decrease slightly for rental properties near the boundary, staying on the market about one week longer.

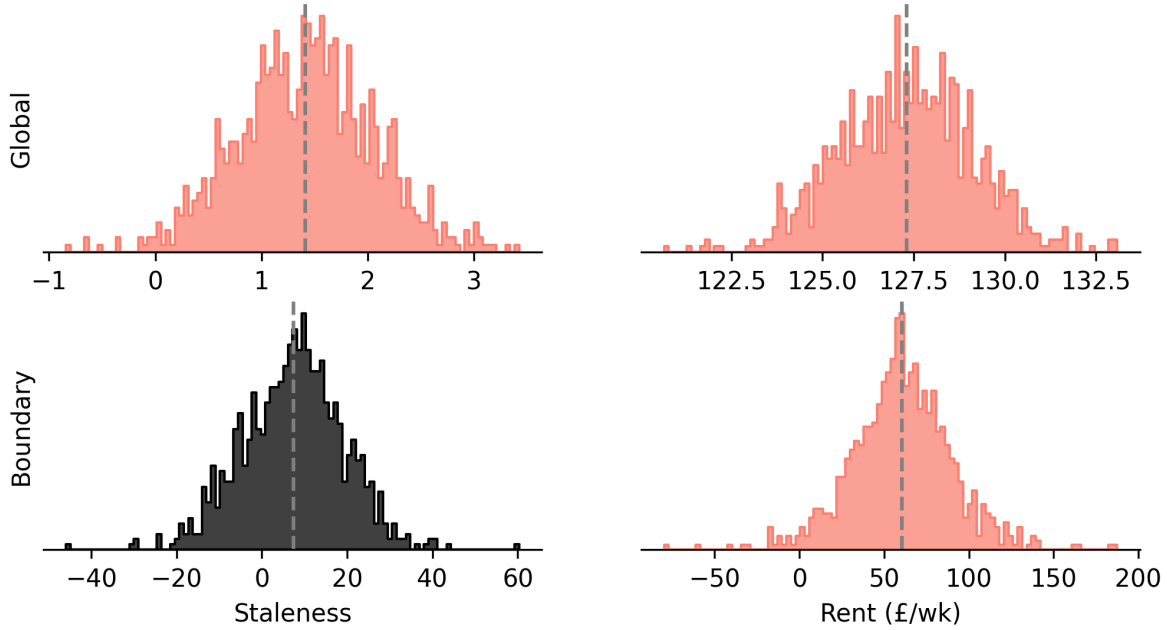


Figure 10: Average Treatment Effect (ATE)s of DLA introduction and distribution of bootstrapped ATEs to indicate variability.

X-learner Results

Similar results manifest from the X-learner identification strategy. In this case, we fit local models inside of and outside of the DLA, including the quartet predecessor outcome as a predictor. We then calculate the difference between the actual model and counterfactual model at the boundary, and calculate the propensity-weighted average³ of the estimated individual treatment effects. Considering model adequacy in Figure 11, we see quite high accuracy again.

Finally, we consider the causal effect estimates in Figure 12 and Table 5. In this case, it again appears that there is a slight increase in rent-per-week due in regulated areas than outside regulated areas, since the treatment effect is slightly positive. In practice, this means we see about a £12/week increase to rent levels within the DLA, even after taking into account starting rent levels and price fluctuations due to COVID-19.⁴ This is lower than the SDID estimate, but still plausible within the distribution of SDID residual bootstraps shown in Figure 10. This effect is also apparent in the total market plots shown in Figure 2 and Figure 4, but now localized directly to the listings affected by the DLA or their substitutes. In practice, this means a tenant might expect to pay around £50 more per month for a DLA-let property.

³The propensity model simply scores whether or not an individual listing is likely to enter the treatment set. This is useful when there is heterogeneity in the treated and untreated population—for example, very large properties may exist among untreated listings

⁴Variability in these plots is likely understated, so claims of significance are not made here.

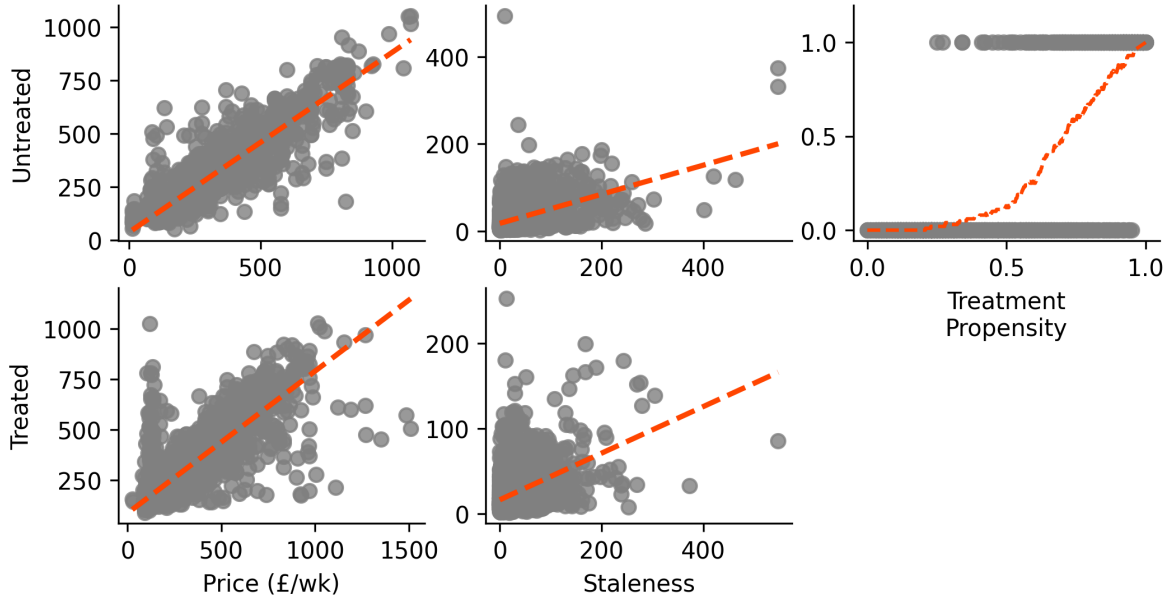


Figure 11: Model adequacy plots for the X-Learner identification strategy. Note the poor predictive accuracy for the listing duration outcome, which affects all subsequent interpretations.

Likewise, clearing rates drop dramatically within the DLA, with staleness declining by an average of 123 days in this model, but given the weak model performance for this outcome, this difference is unlikely to be real.

Table 5: X-learner estimated ATE

		Price	Duration
0	ATE	12.06	-123.02

Discussion and Conclusion

Discretionary licensing authority is an important aspect of local government authority to regulate the private rental sector in the UK. While the use of these policies must generally target concerns about poor housing quality, urban market failure, anti-social behavior, or overcrowding, some argue that the use of selective and additional licensing authority introduces unnecessary additional costs (both monetary and non-monetary) that get passed off onto tenants through higher rents. In addition, these policies have been suggested to drive landlords out of the rental market.

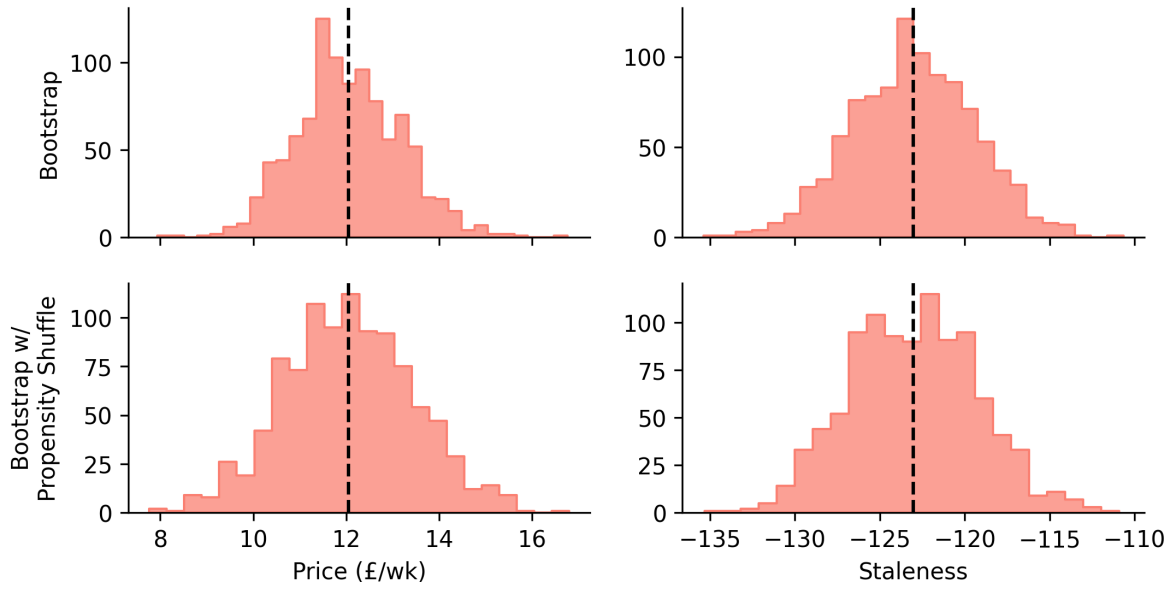


Figure 12: Similar to Figure 10, this shows the distribution of bootstrapped ATEs, alongside the estimated effect. Propensity-shuffled estimates analyse the possibility that the propensity model is misspecified by simply shuffling the propensity scores assigned by the model at random.

I find only slight evidence of reduced supply after the introduction of discretionary licensing policies in central Bristol wards starting in 2019. Indeed, rent growth is slightly higher just inside of the DLA boundaries. More worryingly, rental market clearing appears to have rapidly increased, with the average listing appearing for a much shorter time after 2019. This could be for two reasons—either the clearing rates are actually significantly increased, and properties are quickly being claimed by prospective tenants willing to pay marginally higher rents, or the DLA regulation changes the incentives of landlords and agents with respect to professional management of rental listings. The former is much more likely (Marsh et al. 2023). Further, a tight clearing time for listings removes one avenue for rents going *down*, as agents and/or landlords of stale listings reduce the rent to find tenants at lower price points. This suggests that supply did tighten after the introduction of the DLA, potentially even within the year between the introduction of the city center DLA and COVID-19. Despite controls and data selection strategies to avoid for later COVID-19 effects on housing preferences, properties at the boundary of the city center DLA boundary tended to grow in rent more quickly than properties outside of the DLA boundary. over 2018-2021. On average over all properties at the boundary, this corresponded to about £12/week higher rent than comparator properties at the boundary.

Identifying these short-run changes is important for building longer-term understanding of the efficacy of these minimum standards-type rental regulations. Indeed, while the impact of price controls on housing quality are well understood for rent *control*, these kinds of minimum standard or inspection-type licensing strategies are not as well understood. Thus, this evidence of higher rents and faster clearing times within DLA boundaries should be replicated elsewhere.

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